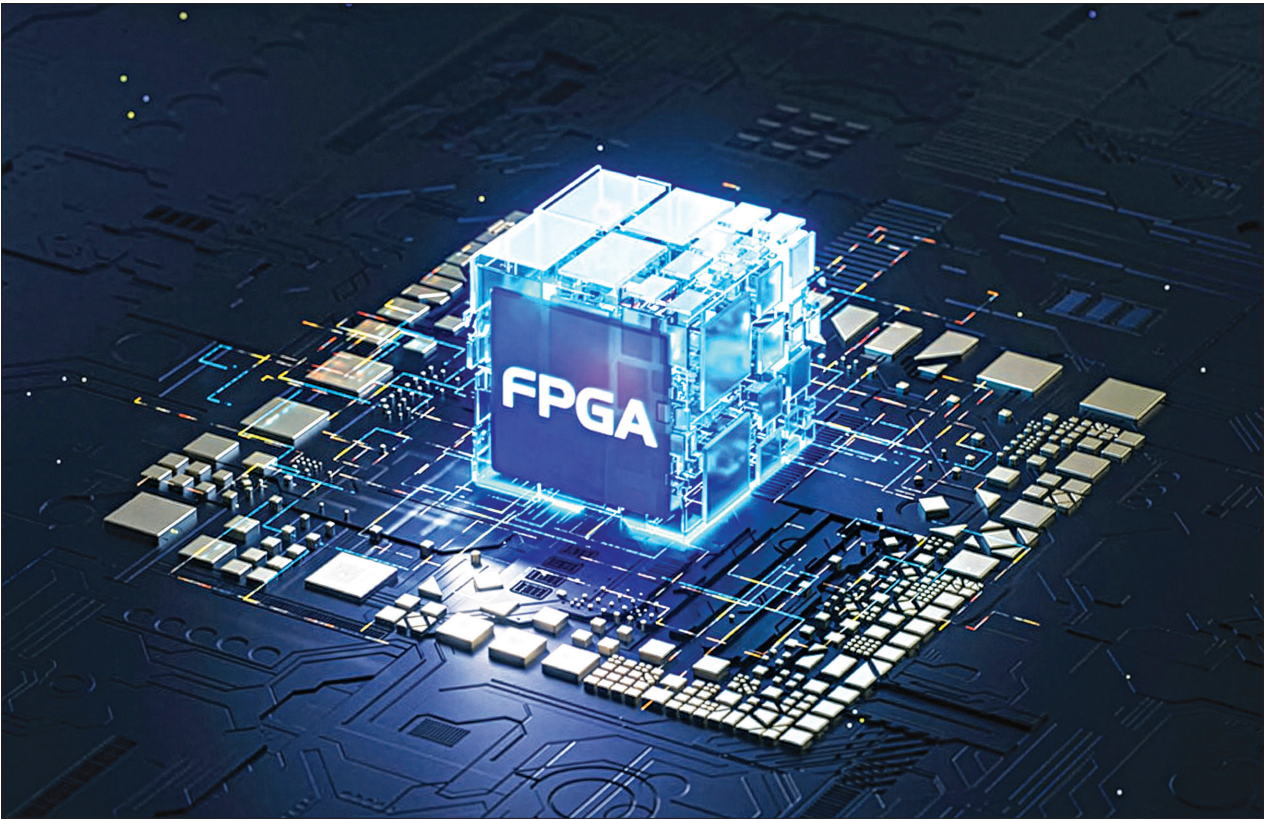




FPGA 101 with open tools

Building your first real bitstream

If the last chapter was where you learned to write hardware, this chapter is where you learn to wear it like a wristwatch – you build a bitstream, load it onto a real FPGA, and watch electrons obey your RTL.

FPGAs are the most addicting “bridge drug” in semiconductors: not as slow and abstract as pure simulation, not as expensive and high-stakes as an ASIC tapeout. You get the dopamine hit of real hardware in minutes, with the freedom to rewrite your “chip” as many times as you like.

This chapter is your practical on-ramp for:

- FPGA basics (what’s inside the chip, without drowning you)

- Picking a beginner-friendly board (and what matters more than marketing)
 - The open-source toolchain (Yosys → nextpnr → bitstream)
 - Constraints (pin mapping: the #1 thing that breaks first projects)
 - A full “Hello FPGA” project: RTL → synth → place&route → bitstream → upload
 - Debug and troubleshooting that actually saves you hours
- We’ll focus on two open-source-friendly FPGA families:
- Lattice iCE40 (Project IceStorm ecosystem; very beginner-friendly)